Zhangsong Li

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Education

Peking University 2023–2028 (expected)
Ph.D. student in Probability
Advisor: Jian Ding

Peking University 2019–2023
B.S. in Mathematics

Research interests

Probability, statistical physics, statistical inference, and theoretical computer science.

Articles

Journal Publication

(J1) **A Polynomial-Time Iterative Algorithm for Random Graph Matching with Non-vanishing Correlation**
Jian Ding, Zhangsong Li
Mathematics of Operations Research, to appear

(J2) **Robust Random Graph Matching in Dense Graphs via an Approximate Message Passing Type Algorithm**
Zhangsong Li
IEEE Transactions on Information Theory, 72(8):6093–6111, 2026

(J3) **Asymptotic Diameter of Preferential Attachment Model**
Hang Du, Shuyang Gong, Zhangsong Li, Haodong Zhu
Electronic Communications in Probability, 31:1–12, 2026

(J4) **A Computational Transition for Detecting Multivariate Shuffled Linear Regression by Low-Degree Polynomials**
Zhangsong Li
IEEE Transactions on Information Theory, 72(4):2444–2456, 2026

(J5) **A Computational Transition for Detecting Correlated Stochastic Block Models by Low-Degree Polynomials**
Guanyi Chen, Jian Ding, Shuyang Gong, Zhangsong Li
Annals of Statistics, 54(1):226–251, 2026

(J6) **Low-Degree Hardness of Detection for Correlated Erdős-Rényi Graphs**
Jian Ding, Hang Du, Zhangsong Li
Annals of Statistics, 53(5):1833–1856, 2025

(J7) **A Polynomial Time Iterative Algorithm for Matching Correlated Gaussian Matrices with Non-vanishing Correlation**
Jian Ding, Zhangsong Li
Foundations of Computational Mathematics, 25(4):1287–1344, 2025

Conference Publication

(C1) **Detection and Reconstruction of a Random Hypergraph from Noisy Graph Projection**
Shuyang Gong, Zhangsong Li, Qiheng Xu
Proceedings of 59th ISIT, pages 1–6. IEEE, 2026

(C2) **Detecting Correlation Efficiently in Very Supercritical Stochastic Block Models: Breaking the Otter's Threshold Barrier**
Guanyi Chen, Jian Ding, Shuyang Gong, Zhangsong Li
Proceedings of 37th SODA, pages 2743–2759. SIAM, 2026

(C3) **Algorithmic Contiguity from Low-Degree Conjecture and Applications in Correlated Random Graphs**
Zhangsong Li
Proceedings of 29th APPROX/RANDOM, pages 30:1–30:18. Schloss Dagstuhl Leibniz-Zentrum für Informatik, 2025

(C4) **Robust Random Graph Matching in Gaussian Models via Vector Approximate Message Passing**
Zhangsong Li
Proceedings of 38th COLT, pages 3580–3581. PMLR, 2025
Conference version of (J2)

Preprints

- **Near Linear Mixing of Random-Cluster Glauber Dynamics on Random Regular Graphs**
  Zhangsong Li
  Preprint, *https://arxiv.org/abs/2611.05794*
- **Free Energy Fluctuations of Symmetric Perceptron Models Up to and Including Criticality**
  Hang Du, Shuyang Gong, Brice Huang, Zhangsong Li
  Preprint, *https://arxiv.org/abs/2610.31736*
- **Improved Rapid Phase Ordering for Ising Dynamics on Random Regular Graphs**
  Zhangsong Li, Allan Sly
  Preprint, *https://arxiv.org/abs/2610.27673*
- **Graph Matching in Random Geometric Graphs I: Efficient Algorithms**
  Dong Huang, Ankang Jiao, Shuangping Li, Zhangsong Li
  Preprint, *https://arxiv.org/abs/2610.16675*
- **Efficient Posterior Sampling for $\mathbb{Z}_2$ Synchronization**
  Zhangsong Li
  Preprint, *https://arxiv.org/abs/2610.08199*
- **Almost Sharp Equivalence between Approximate Message Passing and Low-Degree Polynomials**
  Zhangsong Li
  Preprint, *https://arxiv.org/abs/2609.06988*
- **The Conclave Process**
  Itai Benjamini, Zhenhao Cai, Guanyi Chen, Shuyang Gong, Zhangsong Li

Preprint, *https://arxiv.org/abs/2607.22324*

- **Algorithmic Contiguity from Low-Degree Heuristic II: Predicting Detection-Recovery Gaps**
  Zhangsong Li
  Preprint, *https://arxiv.org/abs/2604.17410*
- **Fundamental Limits of Community Detection in Contextual Multi-Layer Stochastic Block Models**
  Shuyang Gong, Dong Huang, Zhangsong Li
  Preprint, *https://arxiv.org/abs/2602.08173*
- **The Algorithmic Phase Transition in Correlated Spiked Models**
  Zhangsong Li
  Preprint, *https://arxiv.org/abs/2511.06040*
- **Detecting Correlation Efficiently in Stochastic Block Models: Breaking Otter's Threshold in the Entire Supercritical Regime**
  Guanyi Chen, Jian Ding, Shuyang Gong, Zhangsong Li
  Preprint, *https://arxiv.org/abs/2503.06464*
- **A Smooth Computational Transition in Tensor PCA**
  Zhangsong Li
  Preprint, *https://arxiv.org/abs/2509.09904*
- **The Umeyama Algorithm for Matching Correlated Gaussian Geometric Models in the Low-Dimensional Regime**
  Shuyang Gong, Zhangsong Li
  Preprint, *https://arxiv.org/abs/2402.15095*

Teaching and Grading

Fall 2026	TA, Advanced Calculus (I)	Peking University
Spring 2025	TA, Probability Theory	Peking University
Fall 2024	TA, Advanced Probability Theory	Peking University
Fall 2023	TA, Applied Stochastic Process (Honor)	Peking University

Service

- **Assistant in Undergraduate Research Mentorship:**

  (1) Peking University undergraduate research program 2024-2025

  Student: Chenxu Feng (mentored by Jian Ding)

  Title: "Strong Detection Threshold in Correlated Erdős-Rényi Graphs with Constant Average Degree"

  (2) Peking University undergraduate research program 2025-2026

  Students: Chenxu Feng and Yifan Li (mentored by Jian Ding)

  Title: "Structural Properties of the Geometric Preferential Attachment Model"
- **Journal Reviewing:** *Annals of Applied Probability*, *Annals of Statistics*, *Bernoulli*, *IEEE Transactions on Information Theory*, *Information and Inference: A Journal of the IMA*
- **Conference Reviewing:** *SODA 2027*

Talks

The 4th Joint Conference on Statistics and Data Science, Fundamental Limits of Community Detection in Contextual Multi-Layer Stochastic Block Models, July 2026.

The 59th IEEE International Symposium on Information Theory, Detection and Reconstruction of a Random Hypergraph from Noisy Graph Projection, July 2026.

AMSS Stochastic Analysis Seminar (Chinese Academy of Sciences), Recent Progress on Tensor PCA: From Kikuchi Spectral Algorithm to Tensor Networks, June 2026.

Peking University Sino-Russian Student Mathematical Seminar, The Algorithmic Phase Transition for Correlated Spiked Models, December 2025.

The 29th International Conference on Randomization and Computation, Algorithmic Contiguity from Low-Degree Conjecture and Applications in Correlated Random Graphs, August 2025.

The 38th Annual Conference on Learning Theory, Robust Random Graph Matching in Gaussian Models via Vector Approximate Message Passing, July 2025.

International Conference on Applied Probability, Robust Random Graph Matching in Gaussian Models via Vector Approximate Message Passing, June 2025.

YMSC Probability Seminar, Asymptotic Diameter of Preferential Attachment Model (joint with Shuyang Gong), May 2025.

Tsinghua University Statistics Seminar, Recent Progress on Random Graph Matching Problems, March 2025.

Tsinghua Sanya International Mathematics Forum, Low-Degree Hardness of Detection for Correlated Erdős-Rényi Graphs, January 2024.

Peking University Elite Training Program, A Polynomial-Time Iterative Algorithm for Random Graph Matching with Non-vanishing Correlation, June 2023.

Selected Honors and Awards

Presidential Scholarship, Peking University, 2023, 2025.

Elite Program in Applied Mathematics, Peking University, 2022–2023.

Gold medal in the 33rd China Mathematical Olympiad, 2018.

Skills

C++, Mathematica, Matlab.

Last updated: October 2026